1B.1 Velocity profiles and the stress components τ_{ij}. For each of the following velocity distributions, draw a meaningful sketch showing the flow pattern. Then find all the components of τ and ρvv for the Newtonian fluid. The parameter b is a constant.

(a) $v_x = by, v_y = 0, v_z = 0$

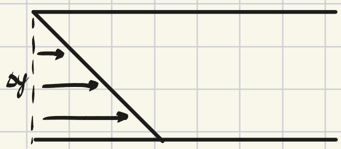
(b) $v_x = by, v_y = bx, v_z = 0$

(c) $v_x = -by, v_y = bx, v_z = 0$

(d) $v_x = -\frac{1}{2}bx, v_y = -\frac{1}{2}by, v_z = bz$

1B-1a - $v_x = by$, $v_y = 0$, $v_z = 0$

$v = [v_x, v_y, v_z]$



$$\tau = \mu(\nabla v + (\nabla v)^{+}) + \lambda(\nabla \cdot v)I = -\mu\begin{bmatrix} 0 & b & 0 \\ b & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = -\mu(b\delta_x\delta_y + b\delta_y\delta_x)$$

$$\nabla v = \begin{bmatrix} \cancel{v_{xx}} & v_{xy} & \cancel{v_{xz}} \\ \cancel{v_{yx}} & \cancel{v_{yy}} & \cancel{v_{yz}} \\ \cancel{v_{zx}} & \cancel{v_{zy}} & \cancel{v_{zz}} \end{bmatrix} = \begin{bmatrix} 0 & \frac{\partial v_x}{\partial y} & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & b & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\nabla \cdot v = \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z} = 0 + 0 + 0 = 0 \qquad \nabla v^{+} = \begin{bmatrix} 0 & 0 & 0 \\ b & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$v = [by, 0, 0]$

$$\rho v v = \begin{bmatrix} v_x \\ v_y \\ v_z \end{bmatrix}\begin{bmatrix} v_x & v_y & v_z \end{bmatrix}\cdot\rho = \begin{bmatrix} \rho v_x^2 & \cancel{\rho v_{xy}} & \cancel{\rho v_{xz}} \\ \cancel{\rho v_{yx}} & \cancel{\rho v_y^2} & \cancel{\rho v_{yz}} \\ \cancel{\rho v_{zx}} & \cancel{\rho v_{zy}} & \cancel{\rho v_z^2} \end{bmatrix} = \rho(by)^2 = \rho b^2 y^2 \delta_x\delta_x$$

1B-1b

$v_x = by$ $v_y = bx$ $v_z = 0$

$v = [by, bx, 0]$

$$\tau = \mu(\nabla v + (\nabla v)^{+}) + \lambda(\nabla \cdot v)I = -\mu\begin{bmatrix} 0 & 2b & 0 \\ 2b & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\nabla v = \begin{bmatrix} \cancel{v_{xx}} & v_{xy} & \cancel{v_{xz}} \\ v_{yx} & \cancel{v_{yy}} & \cancel{v_{yz}} \\ \cancel{v_{zx}} & \cancel{v_{zy}} & \cancel{v_{zz}} \end{bmatrix} = \begin{bmatrix} 0 & b & 0 \\ b & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{\nabla v^{+}} \begin{bmatrix} 0 & b & 0 \\ b & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\nabla \cdot v = \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z} = 0 + 0 + 0 = 0$$

$$\rho v v = \begin{bmatrix} by \\ bx \\ 0 \end{bmatrix}\begin{bmatrix} by & bx & 0 \end{bmatrix}\rho = \rho\begin{bmatrix} b^2y^2 & b^2xy & 0 \\ b^2xy & b^2x^2 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

1B-1c

$v_x = -by$ $v_y = bx$ $v_z = 0$ $v = [-by \;\; bx \;\; 0]$

$$\tau = \mu(\nabla v + (\nabla v)^{+}) + \lambda(\nabla \cdot v)I = -\mu\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = 0$$

$$\nabla v = \begin{bmatrix} \cancel{v_{xx}} & v_{xy} & \cancel{v_{xz}} \\ v_{yx} & \cancel{v_{yy}} & \cancel{v_{yz}} \\ \cancel{v_{zx}} & \cancel{v_{zy}} & \cancel{v_{zz}} \end{bmatrix} = \begin{bmatrix} 0 & -b & 0 \\ b & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{\nabla v^{+}} \begin{bmatrix} 0 & b & 0 \\ -b & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\nabla \cdot v = \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z} = 0 + 0 + 0 = 0$$

$$\rho v v = \begin{bmatrix} -by \\ bx \\ 0 \end{bmatrix}\begin{bmatrix} -by & bx & 0 \end{bmatrix}\rho = \rho\begin{bmatrix} b^2y^2 & -b^2xy & 0 \\ -b^2xy & b^2x^2 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

1b-1d $v_x = -\frac{1}{2}bx$ $v_y = -\frac{1}{2}by$ $v_z = bz$

$v = [-\frac{1}{2}bx, -\frac{1}{2}by, bz]$

$$\tau = \mu(\nabla v + (\nabla v)^{+}) + \lambda(\nabla \cdot v)I = -\mu\begin{bmatrix} -b & 0 & 0 \\ 0 & -b & 0 \\ 0 & 0 & 2b \end{bmatrix}$$

$$\nabla v = \begin{bmatrix} v_{xx} & \cancel{v_{xy}} & \cancel{v_{xz}} \\ \cancel{v_{yx}} & v_{yy} & \cancel{v_{yz}} \\ \cancel{v_{zx}} & \cancel{v_{zy}} & v_{zz} \end{bmatrix} = \begin{bmatrix} -\frac{1}{2}b & 0 & 0 \\ 0 & -\frac{1}{2}b & 0 \\ 0 & 0 & b \end{bmatrix} \xrightarrow{\nabla v^{+}} \begin{bmatrix} -\frac{1}{2}b & 0 & 0 \\ 0 & -\frac{1}{2}b & 0 \\ 0 & 0 & b \end{bmatrix}$$

$$\nabla \cdot v = \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z} = -\frac{1}{2}b + -\frac{1}{2}b + b = 0$$

$$\rho v v = \begin{bmatrix} -\frac{1}{2}bx \\ -\frac{1}{2}by \\ bz \end{bmatrix}\begin{bmatrix} -\frac{1}{2}bx & -\frac{1}{2}by & bz \end{bmatrix}\rho = \begin{bmatrix} \frac{1}{4}b^2x^2 + \frac{1}{4}b^2xy + -\frac{1}{2}b^2xz \\ \frac{1}{4}b^2xy + \frac{1}{4}b^2y^2 - \frac{1}{2}b^2yz \\ -\frac{1}{2}b^2xz - \frac{1}{2}b^2xz + b^2z^2 \end{bmatrix} = \begin{bmatrix} xb^2 \\ yb^2 \\ zb^2 \end{bmatrix}\begin{bmatrix} \frac{1}{4}x & \frac{1}{4}y & z \\ \frac{1}{4}x & \frac{1}{4}y & \frac{1}{2}z \\ -\frac{1}{2}x & \frac{1}{2}x & z \end{bmatrix}$$

2. Verify that "momentum per unit area per unit time" has the same dimensions as "force per unit area."
3. Compare and contrast the molecular and convective mechanisms for momentum transport.

2- $\vec{p} = m \cdot \vec{v}$

$$\frac{\frac{p}{m^2}}{s} = \frac{kg \cdot m}{s} \cdot \frac{1}{m^2} = \frac{kg}{m \cdot s} \cdot \frac{1}{s} = \frac{kg}{m \cdot s^2}$$

$F = ma$

$$\frac{F}{m^2} \quad \frac{F = kg\left(\frac{m}{s^2}\right)}{m^2 \quad m^2} = \frac{kg\,m}{s^2} \cdot \frac{1}{m^2} = \frac{kg}{m \cdot s^2}$$

Momentum transport

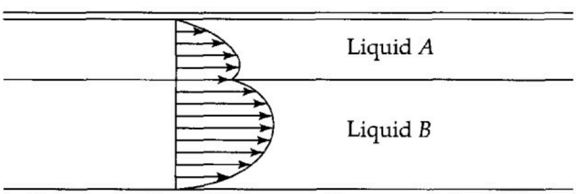
3. Molecular Mechanism
- Occurs by random colisions of molecules in the system
- $\tau_{xy} = -\mu \frac{dv_x}{dy}$

Convective mechanism
- Momentum caried by the fluid
- $\rho v v$

3. How are the Reynolds numbers defined for films, tubes, and spheres? What are the dimensions of Re?

Falling Films $Re = \frac{4\delta \langle v_z \rangle \rho}{\mu}$ tubes $Re = \frac{D \langle v_z \rangle \rho}{\mu}$ $D = 2R$ = tube diameter Spheres $Re = \frac{D v_\infty \rho}{\mu} < 0.1$ Re is dimensionless

11. Two immiscible liquids, *A* and *B*, are flowing in laminar flow between two parallel plates. Is it possible that the velocity profiles would be of the following form? Explain.



- Yes it might be possible for liquid A and liquid B to have the velocity profiles shown above. If the boundary conditions are the same. Such as the top and bottom plates are not moving, and the velocity is continuous at the interface. A difference in viscosities might also explain the difference

2B.2 Alternate procedure for solving flow problems. In this chapter we have used the following procedure: (i) derive an equation for the momentum flux, (ii) integrate this equation, (iii) insert Newton's law to get a first-order differential equation for the velocity, (iv) integrate the latter to get the velocity distribution. Another method is: (i) derive an equation for the momentum flux, (ii) insert Newton's law to get a second-order differential equation for the velocity profile, (iii) integrate the latter to get the velocity distribution. Apply this second method to the falling film problem by substituting Eq. 2.2-14 into Eq. 2.2-10 and continuing as directed until the velocity distribution has been obtained and the integration constants evaluated.

2.2-14 — $\tau_{xz} = -\mu \frac{dv_z}{dx}$

2.2-10 — $\frac{d\tau_{xz}}{dx} = \rho g \cos\beta$

$$\frac{d}{dx}\left(-\mu \frac{dv_z}{dx}\right) = \rho g \cos\beta$$

$$\frac{-\mu \frac{d^2 v_z}{dx^2}}{-\mu} = \frac{\rho g \cos\beta}{-\mu}$$

$$\int \frac{d^2 v_z}{dx^2}\,dx = -\int \frac{\rho g \cos\beta}{\mu}\,dx$$

Boundary conditions: $\tau_{xz} = 0$ when $x = 0$

$-\mu \frac{dv_z}{dx} = 0, \; x = 0$

$-\frac{dv_x}{dx} = 0, \; x = 0$

$v_z = 0, \; x = \delta$

$$\frac{dv_z}{dx} = -\frac{\rho g \cos\beta}{\mu} x + C_1 \implies \int \frac{dv_x}{dx}\,dx = -\int \frac{\rho g \cos\beta\, x}{\mu}\,dx = v_x = -\frac{\rho g \cos\beta\, x^2}{2\mu} + C_2$$

$$0 = -\frac{\rho g \cos\beta}{\mu}(0) + C_1 \Rightarrow C_1 = 0$$

$$0 = \frac{-\rho g \cos\beta\, \delta^2}{2\mu} + C_2$$

$$C_2 = \frac{\rho g \cos\beta\, \delta^2}{2\mu}$$

$$v_x = -\frac{\rho g \cos\beta\, x^2}{2\mu} + \frac{\rho g \cos\beta\, \delta^2}{2\mu}$$

2B.3 Laminar flow in a narrow slit (see Fig. 2B.3).

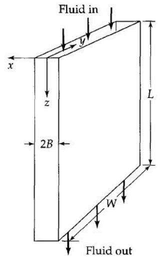


Fig. 2B.3 Flow through a slit, with $B << W << L$.

(a) A Newtonian fluid is in laminar flow in a narrow slit formed by two parallel walls a distance $2B$ apart. It is understood that $B << W$, so that "edge effects" are unimportant. Make a differential momentum balance, and obtain the following expressions for the momentum-flux and velocity distributions:

$$\tau_{xz} = \left(\frac{\mathcal{P}_0 - \mathcal{P}_L}{L}\right)x \tag{2B.3-1}$$

$$v_z = \frac{(\mathcal{P}_0 - \mathcal{P}_L)B^2}{2\mu L}\left[1 - \left(\frac{x}{B}\right)^2\right] \tag{2B.3-2}$$

In these expressions $\mathcal{P} = p + \rho g h = p - \rho g z$.

(b) What is the ratio of the average velocity to the maximum velocity for this flow?

(c) Obtain the slit analog of the Hagen–Poiseuille equation.

(d) Draw a meaningful sketch to show why the above analysis is inapplicable if $B = W$.

(e) How can the result in (b) be obtained from the results of §2.5?

- $v_z = v_z(x)$
- $v_x, v_y = 0$
- $p = \mathcal{P}(x)$

a) z-momentum into shell - $W\Delta x\,\phi_{zz}$, $z=0$

out of shell - $W\Delta x\,\phi_{zz}$, $z=L$

into shell at x - $WL\,\phi_{xz}$, x

out of shell at x+Δx - $WL\,\phi_{xz}$, $x+\Delta x$

Gravitational force on shell - $WL\Delta x\rho g$

Rate of all momentum in - Rate of all momentum out + Force of gravity = 0

$$W\Delta x\,\phi_{zz}\big|_{z=0} - W\Delta x\,\phi_{zz}\big|_{z=L} + WL\,\phi_{xz}\big|_{x} - WL\,\phi_{xz}\big|_{x+\Delta x} + WL\Delta x\,\rho g = 0$$

$$\frac{\cancel{W\Delta x}\,\phi_{zz}|_{z=0}}{W L \cancel{\Delta x}} - \frac{\cancel{W\Delta x}\,\phi_{zz}|_{z=L}}{W L\cancel{\Delta x}} + \frac{\cancel{WL}\,\phi_{xz}|_{x}}{\cancel{WL}\,\Delta x} - \frac{\cancel{WL}\,\phi_{xz}|_{x+\Delta x}}{\cancel{WL}\,\Delta x} + \frac{\cancel{WL\Delta x}\,\rho g}{\cancel{WL\Delta x}} = \frac{0}{WL\Delta x}$$

$$\frac{\phi_{zz}|_{z=0} - \phi_{zz}|_{z=L}}{L} + \lim_{\Delta x\to 0}\frac{\phi_{xz}|_{x} - \phi_{xz}|_{x+\Delta x}}{\Delta x} + \rho g = 0$$

$$\frac{\phi_{zz}|_{z=0} - \phi_{zz}|_{z=L}}{L} - \frac{d\phi_{xz}}{dx} + \rho g = 0$$

$$\phi_{xz} = \tau_{xz} + \cancel{\rho v_x v_z}^{\,0} = \tau_{xz}$$

$$\phi_{zz} = p\delta_{xz} + \cancel{\tau_{zz}}^{\,0} + \rho v_z v_z = \mathcal{P}(z) + \rho v_z^2$$

$$\mathcal{P}(0) + \cancel{\rho v_z^2}^{\,0}\big|_{z=0} - \mathcal{P}(L) + \cancel{\rho v_z^2}^{\,0}\big|_{z=L} - \frac{d\tau_{xz}}{dx} + \rho g = 0$$

$$\frac{\mathcal{P}(0) - \mathcal{P}(L) + \rho g L}{L} - \frac{d\tau_{xz}}{dx} = 0$$

$$\frac{\mathcal{P}(0) - \rho g 0 - (\mathcal{P}(L) - \rho g L)}{L} - \frac{d\tau_{xz}}{dx} = 0$$

$$\tau_{xz} = -\mu\frac{dv_z}{dx} = -\frac{\mathcal{P}_0 - \mathcal{P}_L}{L}x$$

$$\frac{d\tau_{xz}}{dx} = \frac{\mathcal{P}_0 - \mathcal{P}_L}{L}$$

$$\frac{d}{dx}\left(-\mu\frac{dv_z}{dx}\right) = \frac{\mathcal{P}_0 - \mathcal{P}_L}{L}$$

$$v_z = -\frac{\mathcal{P}_0 - \mathcal{P}_L}{2\mu L}x^2 + \frac{\mathcal{P}_0 - \mathcal{P}_L}{2\mu L}B^2$$

$$\int\frac{d^2 v_z}{dx^2}dx = -\int\frac{\mathcal{P}_0 - \mathcal{P}_L}{\mu L}dx$$

$$v_z(\pm B) = -\frac{\mathcal{P}_0 - \mathcal{P}_L}{2\mu L}B^2 + C_2 = 0$$

$$v_z = -\frac{\mathcal{P}_0 - \mathcal{P}_L}{2\mu L}x^2 + C_2 \leftarrow \int\frac{dv_z}{dx^2}dx = -\int\frac{\mathcal{P}_0 - \mathcal{P}_L}{\mu L}x + C_1 \Rightarrow C_1 = 0$$

$$\frac{dv_z}{dx}\Big|_{x=0} = C_1 \Rightarrow C_1 = 0$$

b) $v_z(0) = v_{z,max} = \frac{\mathcal{P}_0 - \mathcal{P}_L}{2\mu L} B^2$

$$\langle v_z \rangle = \frac{1}{A-0} \int v_z \, dA$$

$$= \frac{1}{W(2B)} \int_{-B}^{B} \left(-\frac{\mathcal{P}_0 - \mathcal{P}_L}{2\mu L} x^2 + \frac{\mathcal{P}_0 - \mathcal{P}_L}{2\mu L} B^2 \right) W dx$$

$$= \frac{B}{4\mu} \cdot \frac{\mathcal{P}_0 - \mathcal{P}_L}{L} \int_{-B}^{B} \left[1 - \left(\frac{x}{B}\right)^2 \right] dx$$

$$= \frac{B}{4\mu} \frac{\mathcal{P}_0 - \mathcal{P}_L}{L} \left[x - \frac{x^3}{3B^2} \right] \Big|_{-B}^{B}$$

$$= \frac{B}{4\mu} \frac{\mathcal{P}_0 - \mathcal{P}_L}{L} \cdot \frac{4}{3} B = \frac{\mathcal{P}_0 - \mathcal{P}_L}{L} \cdot \frac{B^2}{3\mu}$$

$$\frac{\langle v_z \rangle}{v_{z,max}} = \frac{\cancel{\frac{\mathcal{P}_0 - \mathcal{P}_L}{L}} \cdot \frac{\cancel{B^2}}{3\cancel{\mu}}}{\cancel{\frac{\mathcal{P}_0 - \mathcal{P}_L}{2\mu L}} \cancel{B^2}} = \frac{\frac{1}{3}}{\frac{1}{2}} = \frac{2}{3}$$

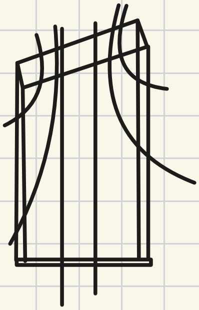
c) $w = \frac{dm}{dt} = \frac{d(\rho V)}{dt} = \rho \frac{dV}{dt}$

$$= \rho \langle v_z \rangle (2BW)$$

$$= \rho \frac{\mathcal{P}_0 - \mathcal{P}_L}{3\mu L} B^2 (2BW)$$

$$= \frac{2}{3} \cdot \frac{(\mathcal{P}_0 - \mathcal{P}_L) B^3 W \rho}{\mu L}$$

d)



e - $v_z^* = \frac{\mathcal{P}_0 - \mathcal{P}_L \, b^2}{2\mu' L} \left[\frac{2\mu'}{\mu' + \mu^2} + \frac{\mu' - \mu^2}{\mu' + \mu^2} \cdot \frac{x}{b} - \left(\frac{x}{b}\right)^2 \right]$

$$\langle v_z' \rangle = \frac{(\mathcal{P}_0 - \mathcal{P}_L)}{12\mu L} b^2 \left(\frac{7\mu' + \mu^2}{\mu' + \mu^2} \right)$$

$$v_z^* = \frac{\mathcal{P}_0 - \mathcal{P}_L \, B^2}{2\mu L} \left[1 - \left(\frac{x}{B}\right)^2 \right]$$

$$= \frac{(\mathcal{P}_0 - \mathcal{P}_L) B^2}{3\mu L}$$

$$v_{max} = v_z'(0) = \frac{(\mathcal{P}_0 - \mathcal{P}_L) B^2}{2\mu L}$$

2B.7 Annular flow with inner cylinder moving axially (see Fig. 2B.7). A cylindrical rod of radius κR moves axially with velocity $v_z = v_0$ along the axis of a cylindrical cavity of radius R as seen in the figure. The pressure at both ends of the cavity is the same, so that the fluid moves through the annular region solely because of the rod motion.

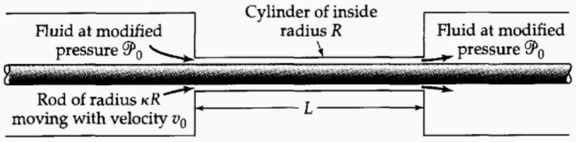


Fig. 2B.7 Annular flow with the inner cylinder moving axially.

(a) Find the velocity distribution in the narrow annular region.
(b) Find the mass rate of flow through the annular region.
(c) Obtain the viscous force acting on the rod over the length L.
(d) Show that the result in (c) can be written as a "plane slit" formula multiplied by a "curvature correction." Problems of this kind arise in studying the performance of wire-coating dies.[1]

b- $w = \frac{dm}{dt} = \frac{d(\rho V)}{dt} = \rho \frac{dV}{dt}$

$= \rho \langle v_z \rangle A$

$= \rho \left(\cancel{\frac{1}{A}} \int_{\kappa R}^{R} v_z dA \right) \cancel{A}$

$= \rho \int_{\kappa R}^{R} v_z (2\pi r\, dr) = 2\pi\rho \int_{\kappa R}^{R} \frac{v_0}{\ln \kappa} r \ln \frac{r}{R} dr$

$u = \frac{r}{R} \longrightarrow r = Ru$

$du = \frac{1}{R} dr \longrightarrow R du = dr$

$= 2\pi\rho \int_{\kappa}^{1} \frac{v_0}{\ln \kappa} (Ru) \ln u (R du)$

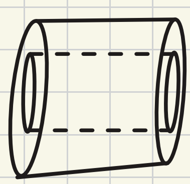
$= \frac{2\pi\rho v_0 R^2}{\ln \kappa} \int_{\kappa}^{1} u \ln u \, du = \frac{2\pi\rho v_0 R^2}{\ln \kappa} \left(-\frac{u^2}{4} + \frac{u^2}{2} \ln u \right) \Big|_{\kappa}^{1}$

$= \frac{\pi R^2 v_0 \rho}{2} \left[\frac{(1-\kappa^2)}{\ln\left(\frac{1}{\kappa}\right)} - 2\kappa^2 \right]$

a) BC. 1 $v_z = v_0$ when $r = \kappa R$

BC. 2 $v_z = 0$ when $r = R$

$p = p(z)$



z-momentum into shell z : $2\pi r \Delta r\, \phi_{zz}|_z$

Out of shell $(z+L)$: $2\pi r \Delta r\, \phi_{zz}|_{z+L}$

into shell at r : $2\pi r L\, \phi_{rz}|_r$

Out of the shell $r+\Delta r$: $[2\pi (r+\Delta r) L\, \phi_{rz}]|_{r+\Delta r}$

gravitational forces on shell : 0

Rate of momentum in $-$ Rate of momentum out $+$ Forces of gravity $= 0$

$\frac{(\cancel{2\pi} r \cancel{\Delta r}) \phi_{zz}|_z}{\cancel{2\pi} \cancel{\Delta r} L} - \frac{(\cancel{2\pi} r \cancel{\Delta r}) \phi_{zz}|_{z+L}}{\cancel{2\pi} \cancel{\Delta r} L} + \frac{(\cancel{2\pi} r \cancel{L}) \phi_{rz}|_r}{\cancel{2\pi} \Delta r \cancel{L}} - \frac{[\cancel{2\pi}(r+\Delta r)\cancel{L}] \phi_{rz}|_{r+\Delta r}}{\cancel{2\pi} \Delta r \cancel{L}} = \frac{0}{2\pi \Delta r L}$

$\frac{r \phi_{zz}|_z}{L} - \frac{r \phi_{zz}|_{z+L}}{L} + \frac{r \phi_{rz}|_r}{\Delta r} - \frac{(r+\Delta r) \phi_{rz}|_r}{\Delta r} = 0$

$-r\left[\frac{\phi_{zz}|_{z+L} - \phi_{zz}|_z}{L}\right] - \lim_{\Delta r \to 0} \left(\frac{(r+\Delta r)\phi_{rz}|_{r+\Delta r} - r\phi_{rz}|_r}{\Delta r} \right) = 0$

$-r\left[\frac{\phi_{zz}|_{z+L} - \phi_{zz}|_z}{L}\right] - \frac{d}{dr}(r\phi_{rz}) = 0$

$\phi_{rz} = \tau_{rz} + \cancel{\rho v_r v_z} = \tau_{rz}$

$\phi_{zz} = p\delta_{zz} + \cancel{\tau_{zz}} + \rho v_z v_z = p + \rho v_z^2$

$-r \frac{\cancel{p|_{z+L}} + \cancel{\rho v_z^2|_{z+L}} - \cancel{p_z} - \rho v_z^2|_z}{L} - \frac{d}{dr}(r\tau_{rz}) = 0$

$\frac{d}{dr}(r\tau_{rz}) = 0$

$\frac{d}{dr}\left(-\mu r \frac{dv_z}{dr}\right) = 0$

$-\mu r \int \frac{d^2 v_z}{dr^2} dr = \int 0\, dr$

$-\mu r \int \frac{dv_z}{dr} = \int c_1 \Rightarrow \int \frac{dv_z}{dr} = -\int \frac{c_1}{\mu r} \Rightarrow v_z = -\frac{c_1}{\mu} \ln r + c_2$

$v_z(\kappa R) = -\frac{c_1}{\mu} \ln(\kappa R) + c_2 = v_0 \Rightarrow -\frac{\mu}{\ln \kappa} v_0$

$v_z(R) = -\frac{c_1}{\mu} \ln R + c_2 = 0 \Rightarrow -\frac{\ln R}{\ln \kappa} v_0$

$\longrightarrow v_z(r) = \frac{1}{\ln \kappa} v_0 \ln r - \frac{\ln R}{\ln \kappa} v_0$

$= \frac{v_0}{\ln \kappa} (\ln r - \ln R)$

$= \frac{v_0}{\ln \kappa} \ln \frac{r}{R}$

$\frac{v_z}{v_0} = \frac{\ln\left(\frac{r}{R}\right)}{\ln \kappa}$

c) $F_z = -\tau_{rz}|_{r=\kappa R} \cdot 2\pi(\kappa R)L$

$\tau_{rz} = -\mu \frac{dv_z}{dr}$

$= -\mu \frac{d}{dr}\left(\frac{v_0}{\ln \kappa} \ln \frac{r}{R} \right)$

$= -\mu \frac{v_0}{\ln \kappa} \frac{\cancel{R}}{r} \cdot \frac{1}{\cancel{R}}$

$= -\frac{\mu v_0}{\ln \kappa} \cdot \frac{1}{r}$

$F_z = -\left(-\frac{\mu v_0}{\ln \kappa} \cdot \frac{1}{r}\right)\Big|_{r=\kappa R} \cdot 2\pi(\kappa R)L$

$= \frac{2\pi \cancel{\kappa R} L \mu_0}{\ln \kappa} \cdot \frac{1}{\cancel{\kappa R}}$

$= \frac{2\pi L \mu v_0}{\ln \kappa} = -\frac{2\pi L \mu v_0}{-\ln \kappa}$

$F_z = -\frac{2\pi L \mu v_0}{\ln(1/\kappa)}$

d) $\kappa = 1 - \varepsilon,\ 0 < \varepsilon \ll 1$

$F_z = -\frac{2\pi L \mu v_0}{\ln(\kappa)}$

$= \frac{2\pi L \mu v_0}{\ln(1-\varepsilon)}$

$= \ln(1-\varepsilon) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} (x-1)^n = -\varepsilon - \frac{\varepsilon^2}{2} - \frac{\varepsilon^3}{3} - \ldots$

$= -\frac{2\pi L \mu v_0}{\varepsilon} \frac{1}{1 + \frac{\varepsilon}{2} + \frac{\varepsilon^2}{3} + \ldots}$

$\tau_{xz} = \frac{\mathscr{P}_0 - \mathscr{P}_L}{L} x - \frac{\mu v_0}{2B}$

2B.8 Analysis of a capillary flowmeter (see Fig. 2B.8). Determine the rate of flow (in lb/hr) through the capillary flow meter shown in the figure. The fluid flowing in the inclined tube is

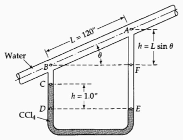


Fig. 2B.8 A capillary flow meter.

[1] J. B. Paton, P. H. Squires, W. H. Darnell, F. M. Cash, and J. F. Carley, *Processing of Thermoplastic Materials*, E. C. Bernhardt (ed.), Reinhold, New York (1959), Chapter 4.

water at 20°C, and the manometer fluid is carbon tetrachloride (CCl_4) with density 1.594 g/cm³. The capillary diameter is 0.010 in. *Note:* Measurements of H and L are sufficient to calculate the flow rate; θ need not be measured. Why?

$v_z = v_z(r)$

z-momentum into shell $z=0$	$2\pi r \Delta r\, \phi_{zz}\big|_{z=0}$
out of shell $z=L$	$2\pi r \Delta r\, \phi_{zz}\big|_{z=L}$
into shell at r	$2\pi(r+\Delta r) L\, \phi_{rz}\big|_{r+\Delta r}$
out of shell @ $r+\Delta r$	$2\pi r \Delta r L\, \rho g \sin\theta$

Momentum in - Momentum out + Gravity = 0

$$\frac{\cancel{2\pi} r \cancel{\Delta r}\, \phi_{zz}|_{z=0}}{\cancel{2\pi \Delta r} L} - \frac{\cancel{2\pi} r \cancel{\Delta r}\, \phi_{zz}|_{z=L}}{\cancel{2\pi \Delta r} L} + \frac{\cancel{2\pi} r \cancel{L}\, \phi_{rz}|_r}{\cancel{2\pi} \Delta r \cancel{L}} - \frac{\cancel{2\pi}(r+\Delta r)\cancel{L}\, \phi_{rz}|_{r+\Delta r}}{\cancel{2\pi} \Delta r \cancel{L}} + \frac{\cancel{2\pi} r \cancel{\Delta r L}\, \rho g \sin\theta}{\cancel{2\pi \Delta r L}} = 0$$

$$\frac{r\phi_{zz}|_{z=0} - r\phi_{zz}|_{z=L}}{L} + \lim_{\Delta r \to 0}\left[\frac{r\phi_{rz}|_r - (r+\Delta r)\phi_{rz}|_{r+\Delta r}}{\Delta r}\right] + r\rho g \sin\theta = 0$$

$$-r\left[\frac{\phi_{zz}|_{z=L} - \phi_{zz}|_{z=0}}{L}\right] - \frac{d}{dr}(r\phi_{rz}) + \rho g r \sin\theta = 0$$

$\phi_{rz} = \tau_{rz} + \cancel{\rho v_r v_z} = \tau_{rz}$

$\phi_{zz} = p\delta_{zz} + \cancel{\tau_{zz}} + \rho v_z v_z = p + \rho v_z^2$

$$-r\left[\frac{p|_{z=L} + \cancel{\rho v_z^2|_{z=L}} - p|_{z=0} - \cancel{\rho v_z^2|_{z=0}}}{L}\right] - \frac{d}{dr}(r\tau_{rz}) + \rho g r \sin\theta = 0$$

$$= -r\left[\frac{p|_{z=L} - p|_{z=0} - \rho g L \sin\theta}{L}\right] - \frac{d}{dr} r\tau_{rz} = 0$$

$$= -r\left[\frac{p|_{z=L} - p|_{z=0} - \rho g h}{L}\right] - \frac{d}{dr} r\tau_{rz} = 0$$

$$\frac{d}{dr} r\tau_{rz} = -r\left[\frac{p|_{z=L} - p|_{z=0} - \rho g h}{L}\right]$$

B.C.1 - $\frac{dv_z}{dr} = 0$, $r = 0$

B.C.2 - $v_z = 0$, $r = R$

$$\int \frac{d}{dr}\left(-\mu r \frac{dv_z}{dr}\right) dr = \int \frac{\rho c - \rho g A}{L} r\, dr$$

$$-\mu r \frac{dv_z}{dr} = \frac{(\rho c - \rho) g A}{2L} r^2 + C_1 \Rightarrow C_1 = 0$$

$$-\mu r \int \frac{dv_z}{dr} dr = \int \frac{(\rho c - \rho) g A}{2L} r\, dr$$

$$v_z = -\frac{(\rho c - \rho) g A}{4\mu L} r^2 + C_2 \Rightarrow C_2 = \frac{(\rho c - \rho) g A}{4\mu L} R^2$$

$$= -\frac{(\rho c - \rho) g A}{4\mu L} r^2 + \frac{(\rho c - \rho) g A}{4\mu L} R^2$$

$$w = \frac{dm}{dt} = \frac{d\rho V}{dt} = \rho \frac{dV}{dt}$$

$$w = \rho \langle v_z \rangle \cdot \pi R^2$$

$$= \rho\left(\frac{1}{\pi R^2}\int v_z\, dA\right)\pi R^2$$

$$= \rho \int_0^R v_z (2\pi r\, dr)$$

$$= 2\pi\rho \int_0^R r v_z\, dr$$

$$2\pi\rho \int_0^R r \frac{\rho c - \rho\; gA}{4\mu L}(R^2 - r^2)\, dr$$

$$= \frac{\pi\rho(\rho c - \rho) g A R^4}{8\mu L}$$